1. F1(a, b, c, d) =m(1, 2, 4, 5, 6, 8, 10, 12, 14)
   F2(a, b, c, d) =m(2, 4, 6, 8, 10, 11, 12, 14, 15)
   Realize F1 and F2 using a PLA. Give the PLA table and internal connection diagram for the PLA.

5. $F_1(a,b,c,d) = \Sigma m(1,2,4,5,6,8,10,12,14)$
$F_2(a,b,c,d) = \Sigma m(2,4,6,8,10,11,12,14,15)$.

ab \ cd	00	01	11	10
00	0	1	0	1
01	1	1	0	1
11	1	0	0	1
10	1	0	0	1

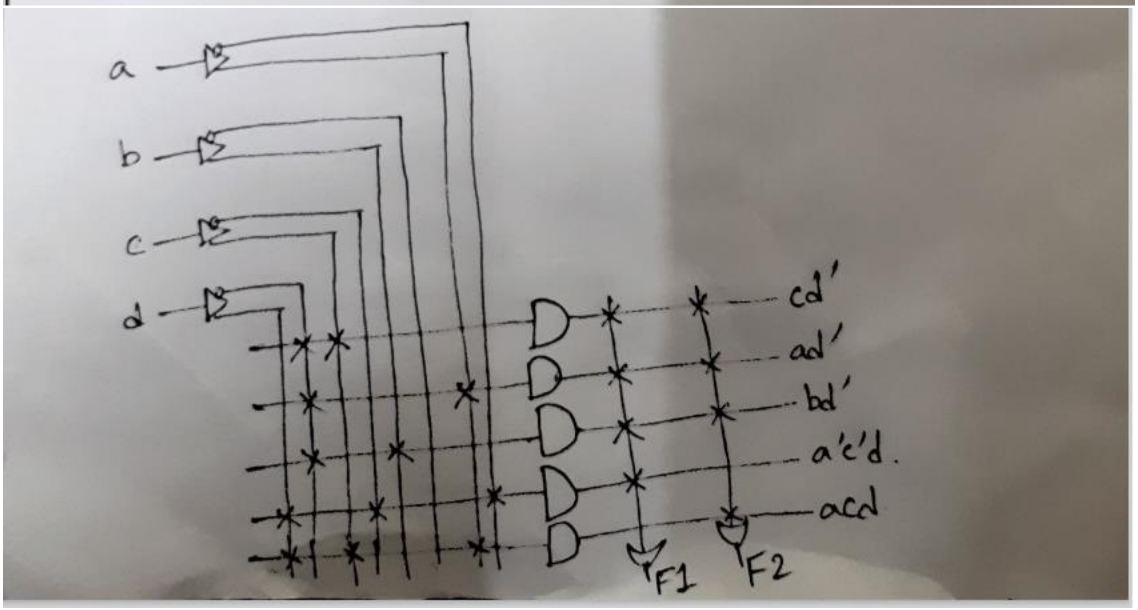
ab \ cd	00	01	11	10
00	0	0	0	1
01	1	0	0	1
11	1	0	1	1
10	1	0	1	1

$F_1 = cd' + a'c'd + ad' + bd'$

$F_2 = cd' + acd + ad' + bd'$

PLA table →

	Product term	Inputs a b c d	Outputs $F_1(T)$ $F_2(T)$
cd'	1	– – 1 0	1 1
ad'	2	1 – – 0	1 1
bd'	3	– 1 – 0	1 1
a'c'd	4	0 – 0 1	1 –
acd	5	1 – 1 1	– 1

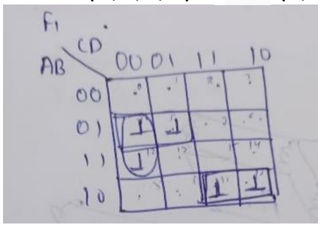


2. Find minimum row PLA table to implement following:
   F1 (A,B,C,D) = Q (0, 1, 2, 3, 6, 7, 8, 9, 13, 14, 15)
   F2 (A,B,C,D) = Q (2, 5, 6, 7, 9, 10, 12, 13, 14, 15)
   F3(A,B,C,D) = Q (1, 2, 3, 5, 6, 7, 8, 9, 11, 13, 15)

Find minimum row PLA table to implement following:

Writing the same in POS form

F1 (A,B,C,D) = ΠM (0, 1, 2, 3, 6, 7, 8, 9, 13, 14, 15) = Σm (4,5,10,11,12)



Similarly for f2,f3:

F2 (A,B,C,D) = ΠM (2, 5, 6, 7, 9, 10, 12, 13, 14, 15) = Σm (0,1,3,4,8,11)

F3 (A,B,C,D) = ΠM (1, 2, 3, 5, 6, 7, 8, 9, 11, 13, 15) = Σm (0,4,10,12,14)

Solution:

using k.map

$f_1 = \overset{(1)}{AB'C} + \overset{(2)}{A'BC'} + \overset{(3)}{BC'D'}$

$f_2 = \overset{(4)}{B'CD} + \overset{(5)}{A'B'C'} + \overset{(6)}{B'C'D'} + \overset{(7)}{A'C'D'}$

$f_3 = \underset{(8)}{ACD'} + \underset{(7)}{A'C'D'} + \underset{(3)}{BC'D'}$

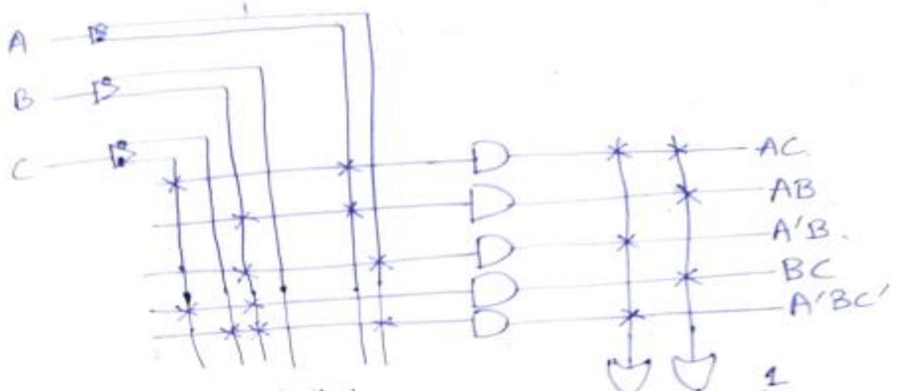
→ total number of terms are 10 but 8 are unique terms. PLA needs only unique terms.

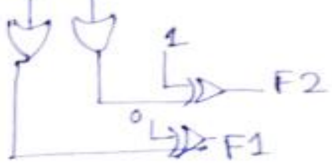
PLA table –

	Product terms	A	B	C	D	F1	F2	F3
1.	AB'C	1	0	1	-	1	-	-
2.	A'BC'	0	1	0	-	1	-	-
3.	BC'D'	-	1	0	0	1	-	1
4.	B'CD	-	0	1	1	-	1	-
5.	A'B'C'	0	0	0	-		1	-
6.	B'C'D'	-	0	0	0	-	1	-
7.	A'C'D'	0	-	0	0	-	1	1
8.	ACD'	1	-	1	0	-	-	1

3. Calculate the number of and, or, xor gates required for implementing a PLA of functions – F1 = A'B + AC + A'BC', F2 = (AC + AB + BC)' If only nand gate is used inside PLA instead of and,or and xor gates calculate the number of total nand gates required in realizing this. If delay of each nand gate is 2 ns, find the time after which outputs will be available.

3. $F1 = A'B + AC + A'BC'$,
$F2 = (AC + AB + BC)'$



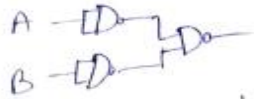


The distinct product terms are → AC, AB, A'B, BC, A'BC'

No of inverters → 3
No. of and gates → 5.
No. of or gates → 2.
No. of xor " → 2.
For inverter, 1 nand gate is required

For and, 2 nand gates are reqd

For or, 3 nand gates are reqd

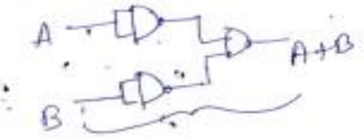


For XOR gate, 4 nand gates are required

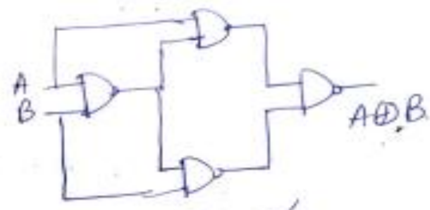
After 2ns , A' is available

Then, each product term like AC, AB etc. will be available after 4ns. ~~if no~~

After 4 ns



After 4ns



After 6ns

So, F1 will be available after → (2+4+4+6) ns

→ 16ns

F2 will be available after → (4+4+6) ns

→ 14ns

As, there are no complemented terms in F2, there is no inverter delay.

4. A circuit has four inputs R,S,T,U and four outputs V,W,Y,Z. RSTU represents a binarycodeddecimal digit. VW represents the quotient and YZ the remainder when RSTU is divided by 3 (VW and YZ represent 2-bit binary numbers). Assume that invalid inputs do not occur. Realize the circuit using a PLA (specify the PLA table).

4. 4

Inputs → R, S, T, U.

O/p's → V, W, Y, Z

VW → quotient, YZ → rem.

3) RSTU (VW

YZ

R S T U	V W Y Z
0 0 0 0	0 0 0 0
0 0 0 1	0 0 0 1
0 0 1 0	0 0 1 0
0 0 1 1	0 1 0 0
0 1 0 0	0 1 0 1
0 1 0 1	0 1 1 0
0 1 1 0	1 0 0 0
0 1 1 1	1 0 0 1
1 0 0 0	1 0 1 0
1 0 0 1	1 1 0 0
1 0 1 0	1 1 0 1
1 0 1 1	1 1 1 0
1 1 0 0	x x x x
1 1 0 1	x x x x
1 1 1 0	x x x x
1 1 1 1	x x x x

RS \ TU	00	01	11	10
00	0	0	0	0
01	0	0	1	1
11	X	X	X	X
10	1	1	1	1

$V = R + ST$

$V' = R'S' + R'T'$

or,

$V' = R'S' + ST'$

RS \ TU	00	01	11	10
00	0	0	1	0
01	1	1	0	0
11	X	X	X	X
10	0	1	1	1

$W = ST' + RU + RT + S'TU$

$W' = ST + R'S'T' + R'TU' + S'T'U'$

RS \ TU	00	01	11	10
00	0	0	0	1
01	0	1	0	0
11	X	X	X	X
10	1	0	1	0

$Y = RT'U' + ST'U + RTU + R'S'TU'$.

$Y' = ST + R'S'T' + RT'U + RTU' + ST'U' + R'TU$

RS \ TU	00	01	11	10
00	0	1	0	0
01	1	0	1	0
11	X	X	X	X
10	0	0	0	1

$Z = ST'U' + STU + RTU' + R'S'T'U$.

$Z' = RT' + RU + R'TU' + S'T'U' + ST'U + S'TU$

And terms → $R, ST, R'S'T', R'TU', S'T'U', RT'U, RTU', ST'U', R'TU, STU, R'S'T'U$

4. PLA table →

	Product term	Inputs R S T U	Outputs V W Y Z
1.	R	1 _ _ _	1 _ _ _
2.	ST	_ 1 1 _	1 1 1 _
3.	R'S'T'	0 0 0 _	_ 1 1 _
4.	R'TU'	0 _ 1 0	_ 1 _ 1
5.	S'T'U'	_ 0 0 0	_ 1 _ 1
6.	RT'U	1 _ 0 1	_ _ 1 _
7.	RTU'	1 _ 1 0	_ _ 1 1
8.	ST'U'	_ 1 0 0	_ _ 1 1
9.	R'TU	0 _ 1 1	_ _ 1 _
10.	STU	_ 1 1 1	_ _ _ 1
11.	R'S'T'U	0 0 0 1	_ _ _ 1

V → True output
W → Complemented o/p
Y → Complemented o/p
Z → True o/p

5. The extra-terrestrial life project team has just discovered aliens living on the bottom of Mono Lake. They need to construct a circuit to classify the aliens by potential planet of origin based on measured features available from the ISRO probe: greenness, brownness, sliminess, and ugliness. Careful consultation with xenobiologists leads to the following conclusions: • If the alien is green and slimy or ugly, brown, and slimy, it might be from Mars. • If the critter is ugly, brown, and slimy, or green and neither ugly nor slimy, it might be from Venus. • If the

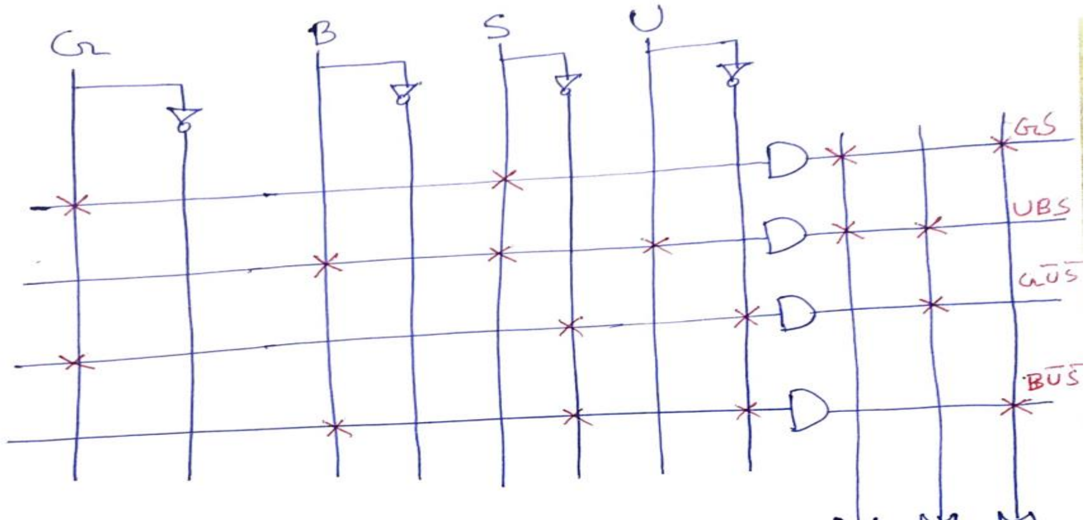
beastie is brown and neither ugly nor slimy or is green and slimy, it might be from Jupiter. Note that this is an inexact science; for example, a life form which is mottled green and brown and is slimy but not ugly might be from either Mars or Jupiter.
Program a 4 × 4 × 3 PLA to identify the alien. You may use dot notation.
4x4x3 denotes inputs,and terms and outputs respectively.

10. Let, Gr denote Green,
B → Brown
S → Sliminess
U → Ugliness.

4 inputs to PLA → Gr, B, S, U.
Outputs → M, V, J.

M → Mars
V → Venus
J → Jupiter.

$M = GrS + UBS$
$V = UBS + Gr\overline{U}\,\overline{S}$
$J = B\overline{U}\,\overline{S} + GrS$



M V J

4x4x3 PLA.
inputs
product terms.
outputs

4 distinct product terms are → GrS, UBS, $Gr\overline{U}\,\overline{S}$, $B\overline{U}\,\overline{S}$.

6. Braille is a system which allows a blind person to read alphanumerics by feeling a pattern of raised dots. Design a circuit that converts BCD to Braille. The table shows the correspon dence between BCD and Braille.By using PLA define the connection pattern.

A	B	C	D	W	X	Z	Y
0	0	0	0		•	•	•
0	0	0	1	•			
0	0	1	0	•		•	
0	0	1	1	•	•		
0	1	0	0	•	•		•
0	1	0	1	•			•
0	1	1	0	•	•	•	
0	1	1	1	•	•	•	•
1	0	0	0	•		•	•
1	0	0	1		•	•	

(1) truth table

A	B	C	D	W	X	Y	Z
0	0	0	0	0	1	1	1
0	0	0	1	1	0	0	0
0	0	1	0	1	0	0	1
0	0	1	1	1	1	0	0
0	1	0	0	1	1	1	0
0	1	0	1	1	0	1	0
0	1	1	0	1	1	0	1
0	1	1	1	1	1	1	1
1	0	0	0	1	0	1	1
1	0	0	1	0	1	0	1
1, 0, 1, 0,				x	x	x	x
1; 1; 1, 1				x	x	x	x

using: k map minimization (rest term as don't care)

$W = A\bar{D} + C + \bar{A}D + B$

for 'W'

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	1	1	1
11	x	x	x	x
10	1	0	x	x

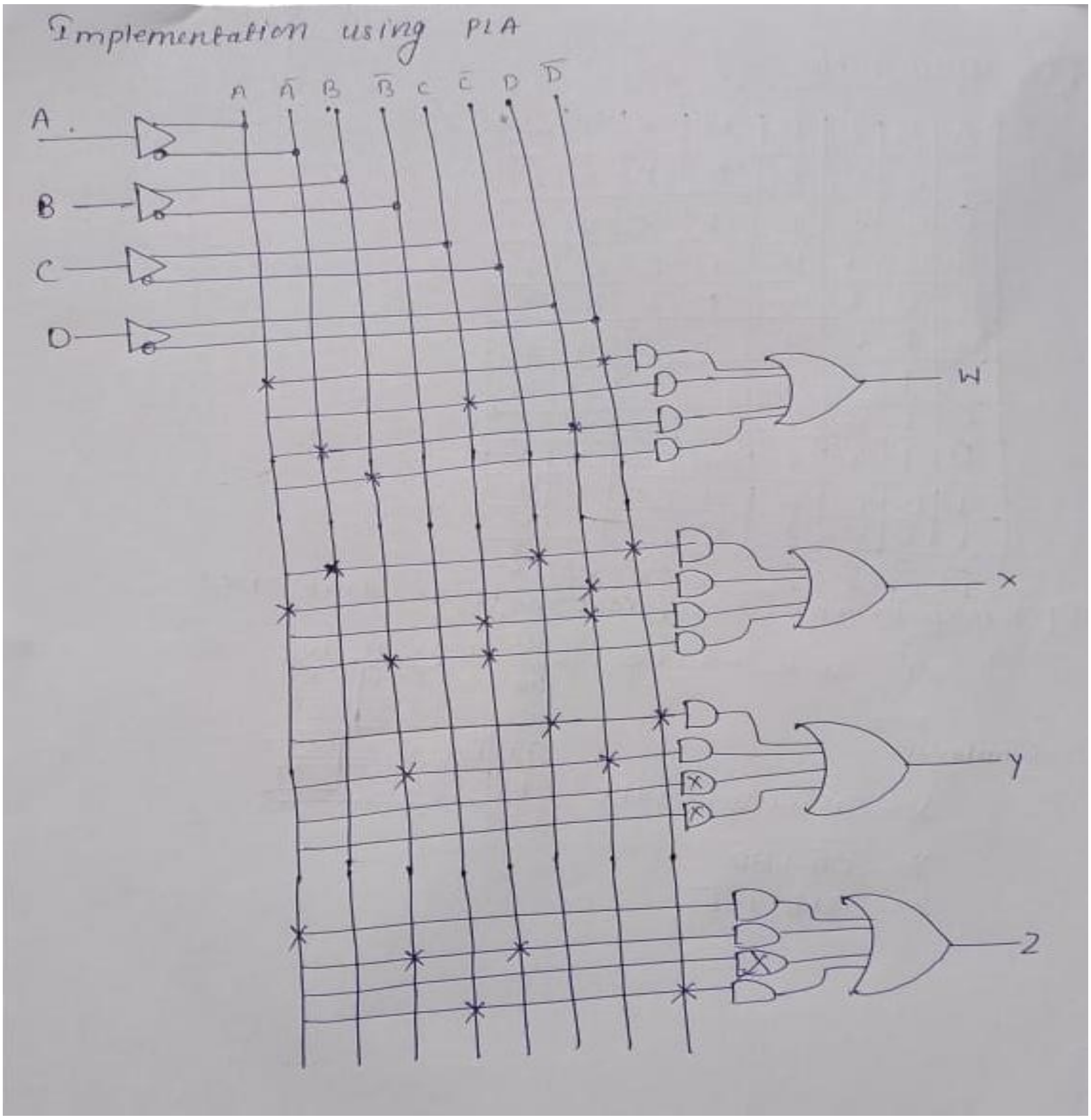
~~x =~~

Similarly

$X = \bar{A}\bar{C}\bar{D} + AD + CD + BC$

$Y = \bar{C}\bar{D} + BD$

$Z = A + BC + \bar{B}\bar{D}$



7. Implement PLA circuit and table for the following equation and repeat the same for PAL

   X= AC’+B

   Y= A’B’+AC’

   Z= B+AC

   Solution:

   For PLA: circuit there is 4 unique terms

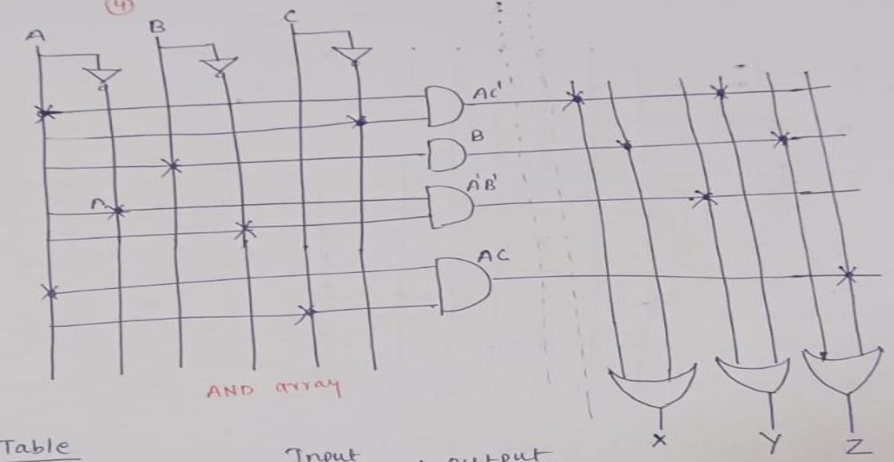
Solution (a)

PLA circuit implementation —

$X = AC' + B$

$Y = A'B' + AC'$

$Z = B + AC$



PLA Table

Product term	Input A	B	C	Output X	Y	Z
AC'	1	-	0	1	1	0
B	-	1	-	1	0	1
A'B'	0	0	-	0	1	0
AC	1	0	1	0	0	1

$X = AC' + B$

$Y = AC' + A'B'$

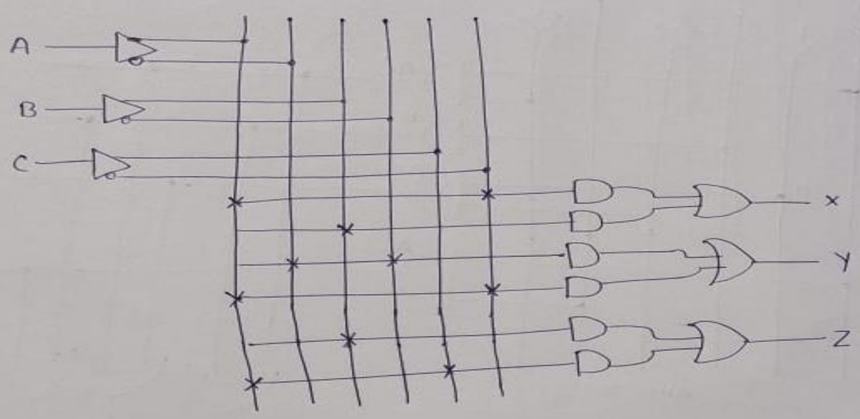
$Z = B + AC$

PAL circuit implementation:

$X = AC' + B$

$Y = A'B' + AC'$

$Z = B + AC$



PAL Table : 6 Product term

Product term	AND inputs A	B	C	Outputs
1	1	[illegible]	0	$X = AC' + B$
2	-	1	-	
3	0	0	-	$Y = A'B' + AC'$
4	1	-	0	
5	-	1	-	$Z = B + AC$
6	1	-	1	

8. Implement a full subtracter using a PAL.Consider difference and borrow out as outputs.

2. Full subtractor truth table →

A	B	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

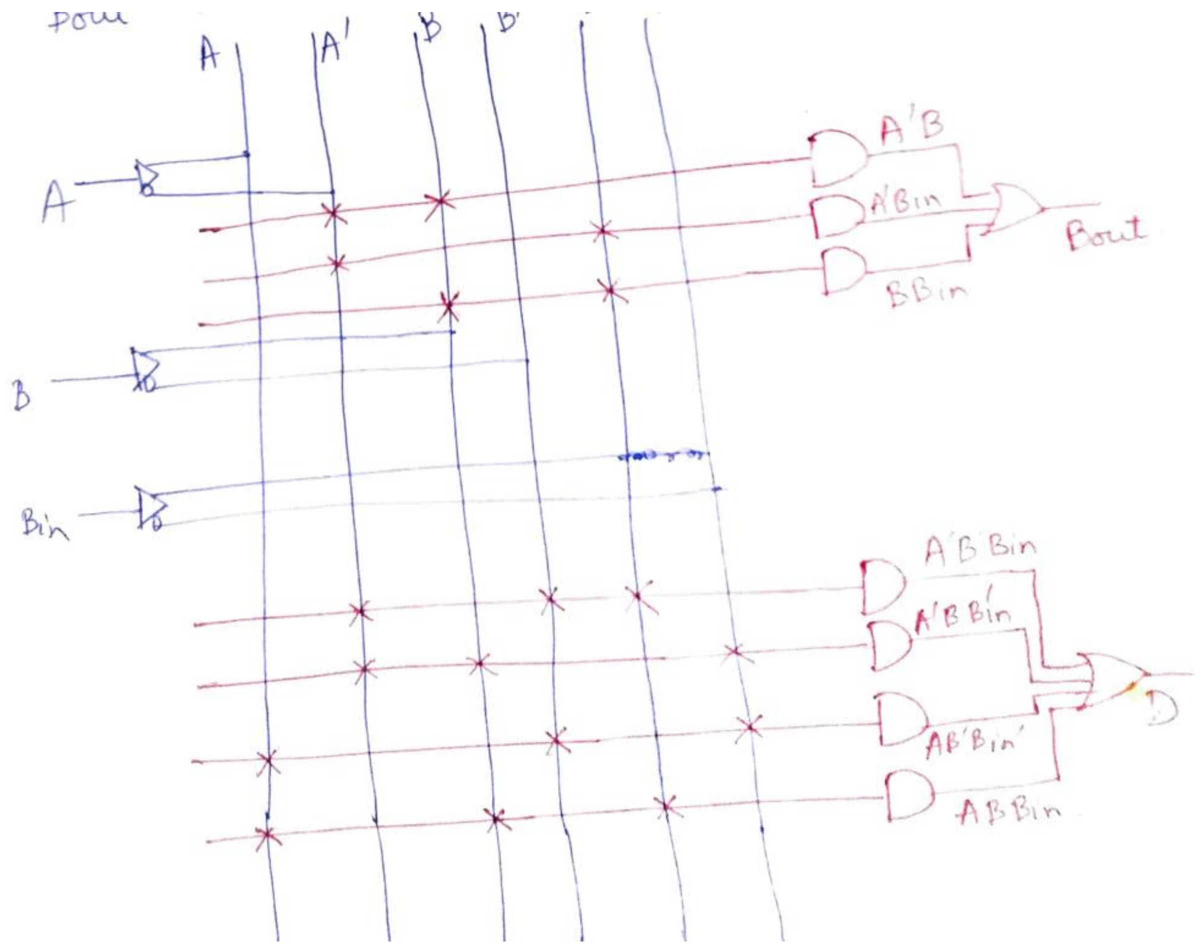
A \ B Bin	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$D = A'B'Bin + A'BBin' + AB'Bin' + ABBin$

A \ B Bin	00	01	11	10
0	0	1	1	1
1	0	0	1	0

$Bout = A'B + A'Bin + BBin$

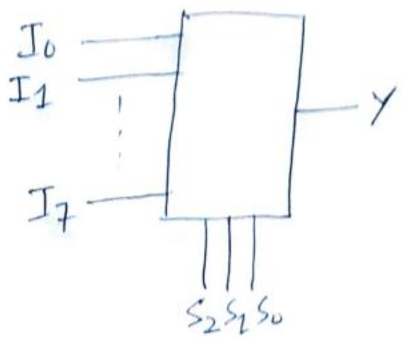
', B, B', Bin, Bin'

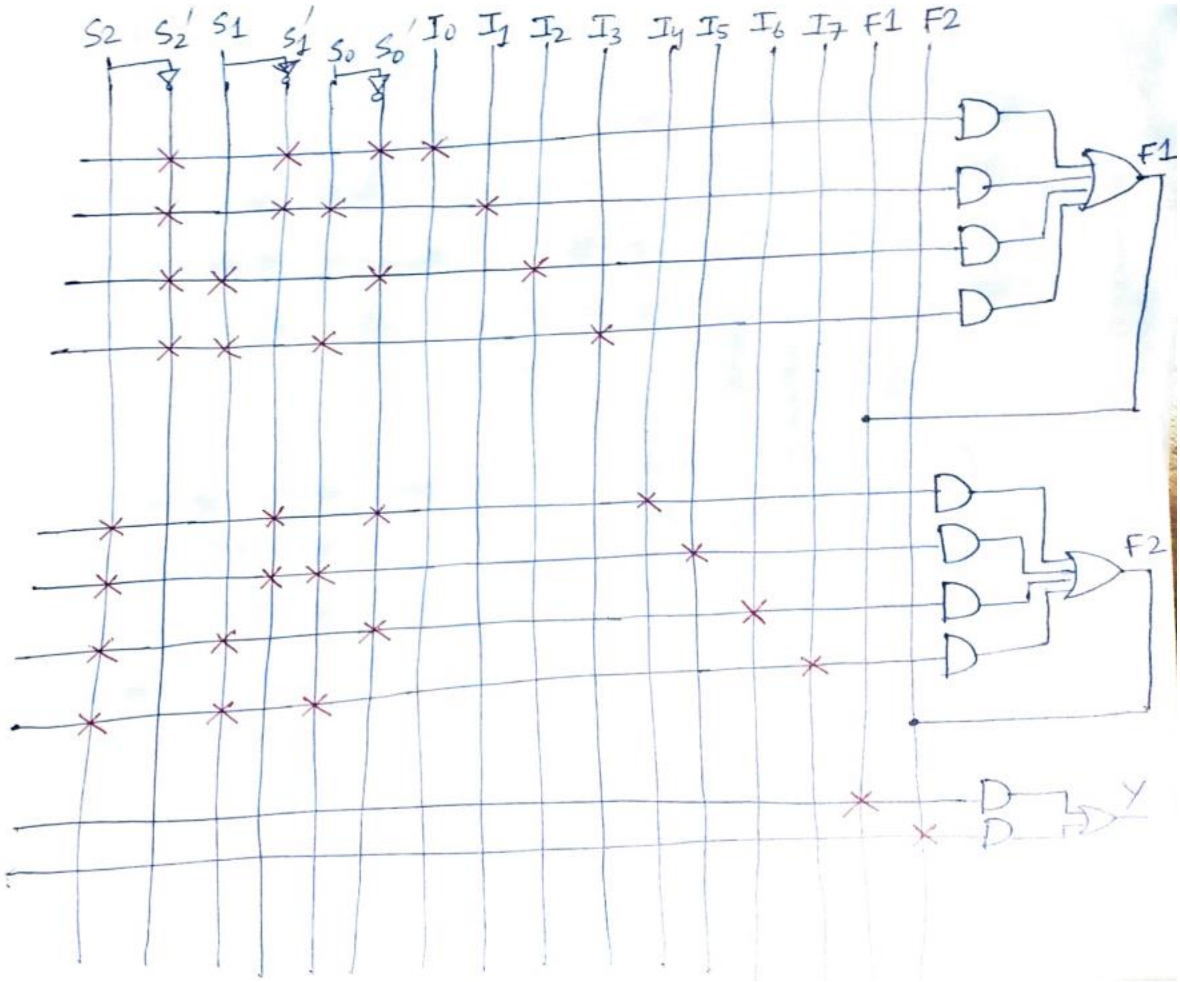


9. a) Show how to make an 8-to-1 MUX using a PAL. Assume that PAL has 14 inputs and six outputs and assume that each output OR gate may have up to four AND terms as inputs.(Hint: Wire some outputs of the PAL around to the inputs, external to the PAL. Some PALs allow this inside the PAL to save inputs.)

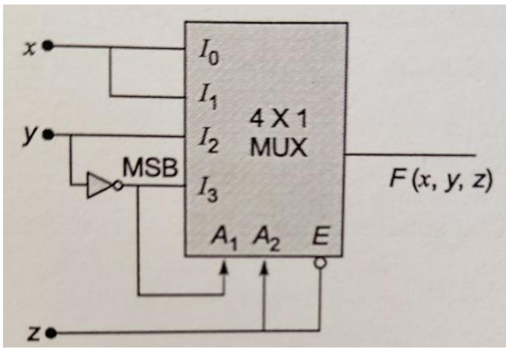
11. 8 to 1 mux using a PAL

$$Y = \bar{S_2}\bar{S_1}\bar{S_0}I_0 + \bar{S_2}\bar{S_1}S_0I_1 + \bar{S_2}S_1\bar{S_0}I_2 + \bar{S_2}S_1S_0I_3 + S_2\bar{S_1}\bar{S_0}I_4 + S_2\bar{S_1}S_0I_5 + S_2S_1\bar{S_0}I_6 + S_2S_1S_0I_7$$





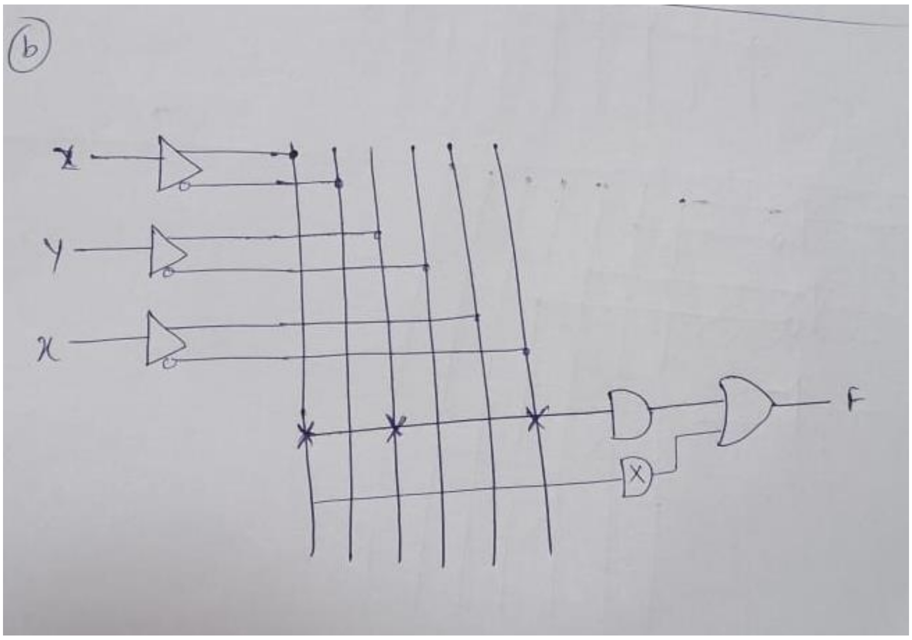
b) Solve the given circuit and implement it by using PAL.Assume that each output OR gate may have up to two AND terms as inputs.



$F=(A'_1A'_2I_0+A'_1A_2I_1+A_1A'_2I_2+A_1A_2I_3)E'$

A1= Y
A2=Z
E=Z
$I_0 = I_1 = X$
$I_2 = Y$
$I_3 = Y'$

So: F= XYZ'



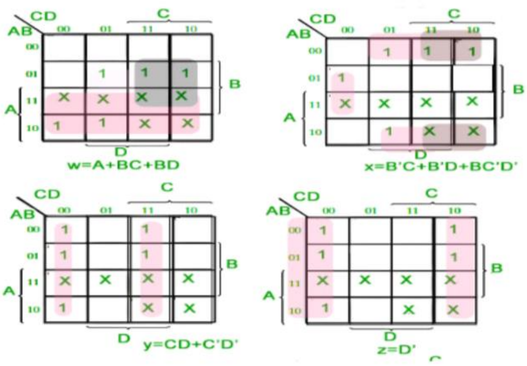
10. List the PAL programming table for the BCD - to Excess-3 converter with the Boolean functions simplified.

Solution:

Step I: Writing BCD to access 3 table(unused terms as don't care)

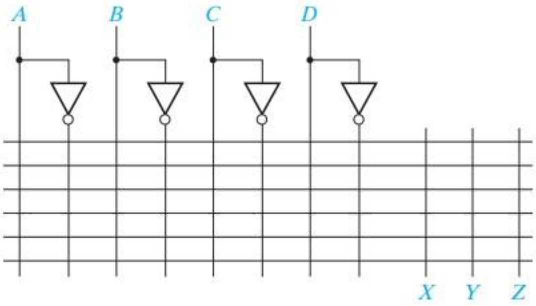
BCD(8421)				Excess-3			
A	B	C	D	w	x	y	z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0
1	0	1	0	X	X	X	X
1	0	1	1	X	X	X	X
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

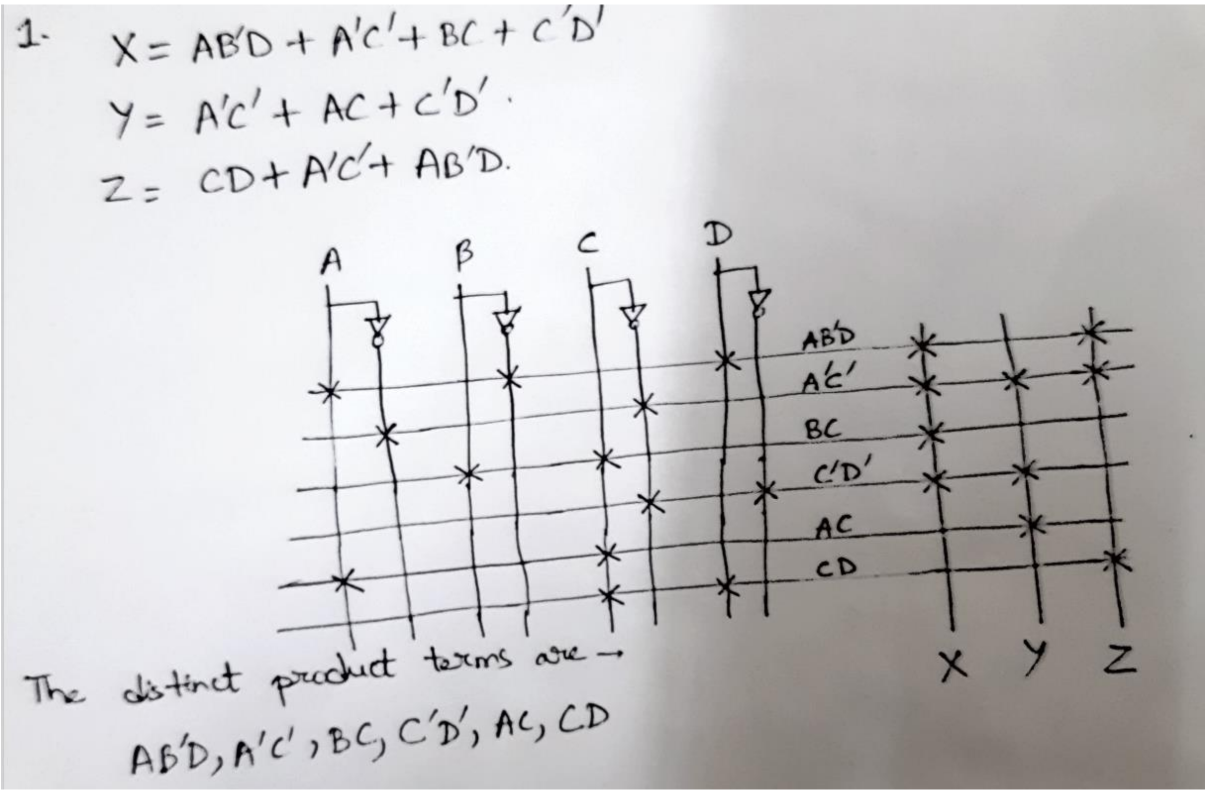
step II : solving equation using K map



Product Term	AND Inputs A	B	C	D	Outputs
1	1	-	-	-	w = A+BC+BD
2	-	1	1	-	
3	-	1	-	1	
4	-	0	1	-	x = B′C + B′D + BC′D′
5	-	0	-	1	
6	-	1	0	0	
7	-	-	1	1	y = CD + C′D′
8	-	-	0	0	
9	-	-	-	-	
10	-	-	-	0	z = D′
11	-	-	-	-	
12	-	-	-	-	

11. The following PLA will be used to implement the following equations: X = AB’D + A’C’ + BC + C’D’ Y = A’C’ + AC + C’D’ Z = CD + A’C’ + AB’D Indicate the connections that will be made to program the PLA to implement these equations.





12. Realize the hexadecimal to ASCII code converter using PLA-

Input $W\ X\ Y\ Z$	Hex Digit	ASCII Code for Hex Digit $A_6\ A_5\ A_4\ A_3\ A_2\ A_1\ A_0$
0 0 0 0	0	0 1 1 0 0 0 0
0 0 0 1	1	0 1 1 0 0 0 1
0 0 1 0	2	0 1 1 0 0 1 0
0 0 1 1	3	0 1 1 0 0 1 1
0 1 0 0	4	0 1 1 0 1 0 0
0 1 0 1	5	0 1 1 0 1 0 1
0 1 1 0	6	0 1 1 0 1 1 0
0 1 1 1	7	0 1 1 0 1 1 1
1 0 0 0	8	0 1 1 1 0 0 0
1 0 0 1	9	0 1 1 1 0 0 1
1 0 1 0	A	1 0 0 0 0 0 1
1 0 1 1	B	1 0 0 0 0 1 0
1 1 0 0	C	1 0 0 0 0 1 1
1 1 0 1	D	1 0 0 0 1 0 0
1 1 1 0	E	1 0 0 0 1 0 1
1 1 1 1	F	1 0 0 0 1 1 0

a. From the given truth table, the outputs are simplified at first by K-map.

WX \ YZ	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	0	1	1

$A_6 = WX + WY$

$A_6' = W' + X'Y'$.

WX \ YZ	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	1	1	0	0

$A_5 = W' + X'Y'$

$A_5' = WX + WY$.

WX \ YZ	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	1	1	0	0

$A_4 = W' + X'Y'$

$A_4' = WX + WY$

WX \ YZ	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	1	1	0	0

$A_3 = WX'Y'$

$A_3' = Y + W' + X$.

WX \ YZ	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	1	1	1
10	0	0	0	0

$A_2 = W'X + XZ + XY$

$A_2' = X' + WY'Z'.$

WX \ YZ	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	1	0
10	0	0	1	0

$A_1 = YZ + W'Y + WXY'Z'$

$A_1' = X'Y' + W'Y' + Y'Z + WYZ'$

WX \ YZ	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	0	0	1
10	0	1	0	1

$A_0 = W'Z + WYZ' + WXZ' + X'Y'Z$

$A_0' = W'Z' + X'Y'Z' + WXZ + WYZ.$

Minimum no. of product terms will be for →

$A_6' = W' + X'Y'$

$A_5 = W' + X'Y'.$

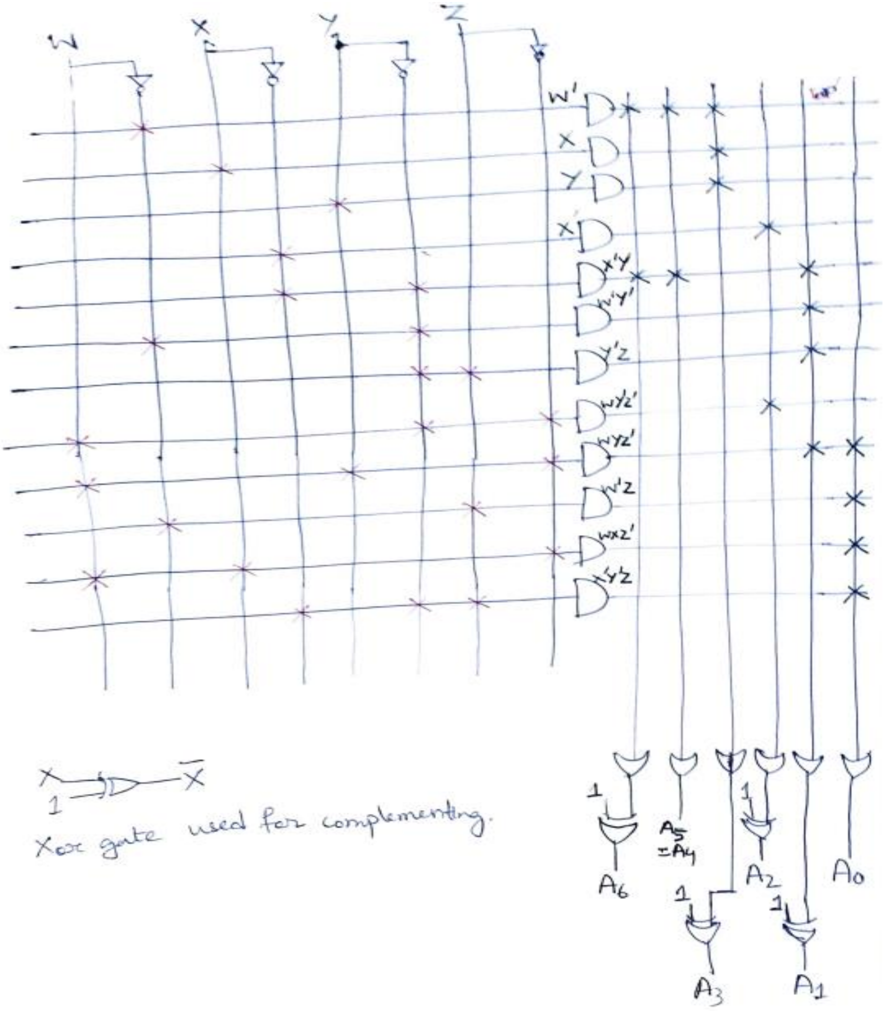
$A_4 = W' + X'Y'.$

$A_3' = W' + X + Y.$

$A_2' = X' + WY'Z'$

$A_1' = X'Y' + W'Y' + Y'Z + WYZ'.$

$A_0 = W'Z + WYZ' + WXZ' + X'Y'Z.$



13. PLA has 3 outputs, F1, F2 and F3. The product terms for F1 are A'B',AC,AB. Product terms for F2 are A'C, AB and F3 are A'B',AC,BC'.

a. What is the minimum number of product terms required to implement F1,F2and F3 using this PLA?

b. What is the minimum number of product terms required to implement F1,F2and F3 using this PAL?

Solution:

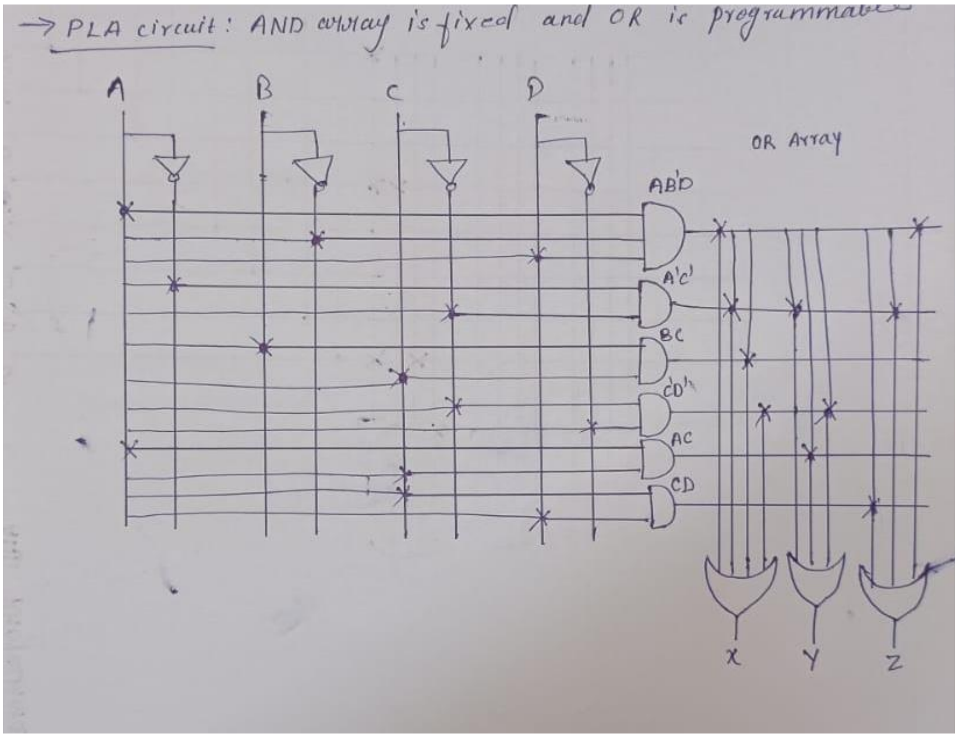
product terms required to implement F1,F2 and F3 using this PLA is 5 (A'B',AC,AB,A'B,BC')

product terms required to implement F1,F2 and F3 using this PAL is 8

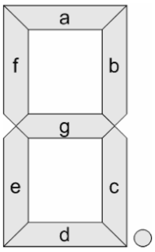
14. . Implement PLA circuit for the following equation

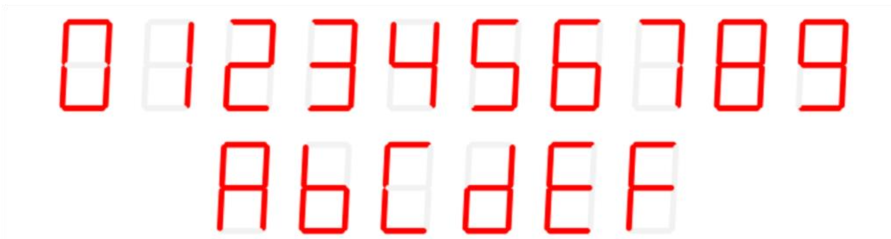
X = AB'D + A'C' + BC + C'D'
Y = A'C' + AC + C'D'
Z = CD + A'C' + AB'D
Solution:

This PLA circuit has **SIX** unique terms AB'D,A'C',BC, C'D', AC , CD



15. Program a PLA to implement a 4-bit binary-to-7-segment decoder. The decoder should display "0" to "9" and "A" to "F" in hexadecimal notation. Show the truth table and the programming equations





Solution-

Inputs					Segments							
A	B	C	D		a	b	c	d	e	f	g	
0	0	0	0		1	1	1	1	1	1	0	For display 0
0	0	0	1		0	1	1	0	0	0	0	For display 1
0	0	1	0		1	1	0	1	1	0	1	For display 2
0	0	1	1		1	1	1	1	0	0	1	For display 3
0	1	0	0		0	1	1	0	0	1	1	For display 4
0	1	0	1		1	0	1	1	0	1	1	For display 5
0	1	1	0		1	0	1	1	1	1	1	For display 6
0	1	1	1		1	1	1	0	0	0	0	For display 7
1	0	0	0		1	1	1	1	1	1	1	For display 8
1	0	0	1		1	1	1	1	0	1	1	For display 9
1	0	1	0		1	1	1	0	1	1	1	For display A
1	0	1	1		0	0	1	1	1	1	1	For display b
1	1	0	0		1	0	0	1	1	1	0	For display C
1	1	0	1		0	1	1	1	1	0	1	For display d
1	1	1	0		1	0	0	1	1	1	1	For display E
1	1	1	1		1	0	0	0	1	1	1	For display F

for segment a(A,B,C,D) = $\Sigma(0,2,3,5,6,7,8,9,10,12,14,15)$

Kmap

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	1	0	1	1
10	1	1	0	1

$a = AB'C' + A'BD + AD' + A'C + BC + B'D'$

$a' = A'B'C'D + A'BC'D' + ABC'D + AB'CD$

1 2 3 4

for segment b(A,B,C,D) = $\Sigma(0,1,2,3,4,7,8,9,10,13)$

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	0	1	0	0
10	1	1	0	1

$b = A'C'D' + A'CD + AC'D + B'C' + B'D'$

$b' = BCD' + ACD + ABD' + A'BC'D$

5 6 7 8

for segment c(A,B,C,D) = $\Sigma(0,1,3,4,5,6,7,8,9,10,11,13)$

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	0	1	0	0
10	1	1	1	1

$c = A'C' + A'D + C'D + A'B + AB'$

$c' = ABC + ABD' + A'B'CD'$

9 7 10

for segment 'd'

$d(A,B,C,D) = \Sigma(0, 2, 3, 5, 6, 8, 9, 11, 12, 13, 14)$

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	1	1	0	1
10	1	1	1	0

$d = A'B'D' + B'CD + BC'D + BCD' + AC'$

$d' = BCD + A'BC'D' + AB'CD' + A'B'C'D$

11 2 12 1

for segment $e(A,B,C,D) = \Sigma(0, 2, 6, 8, 10, 11, 12, 13, 14, 15)$

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	1	1	1	1
10	1	0	1	1

$e = B'D' + CD' + AC + AB$

$e' = AD + A'BC' + B'CD$

13 14 15

for segment $f(A,B,C,D) = \Sigma(0, 4, 5, 6, 8, 9, 10, 11, 12, 14, 15)$

AB \ CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	1	0	1	1
10	1	1	1	1

$f = A'BC' + C'D' + BD' + AB' + AC$

$g' = A'B'D + A'B'C + A'CD + ABC'D$

16 17 18 3

for segment 'g'

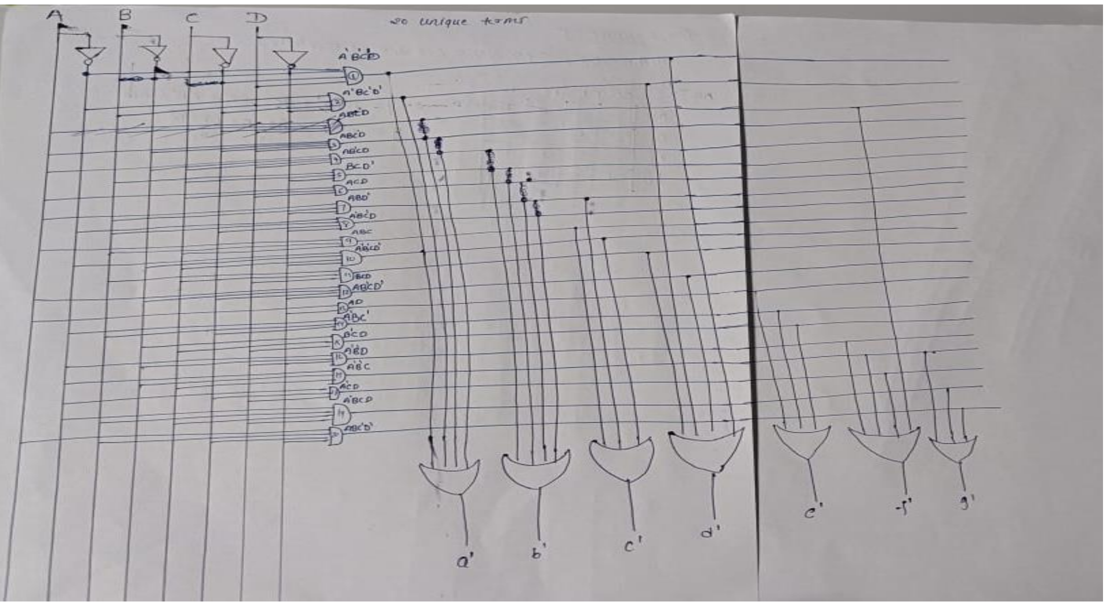
$g(A,B,C,D) = \Sigma(2, 3, 4, 5, 6, 8, 9, 10, 11, 13, 14, 15)$

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	0	1	1	1
10	1	1	1	1

$g = A'BC' + B'C + CD' + AB' + AD$

$g' = A'B'C + A'BCD + ABC'D'$

17 19 20



16. Show how to make an 8-to-3 priority encoder using a PAL. Assume that PAL has 14 inputs and six outputs and assume that each output OR gate may have up to four AND terms as inputs.(Hint: Wire some outputs of the PAL around to the inputs, external to the PAL. Some PALs allow this inside the PAL to save inputs.)

12. The truth table of 8 to 3 priority encoder →

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Q_2	Q_1	Q_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	x	0	0	1
0	0	0	0	0	1	x	x	0	1	0
0	0	0	0	1	x	x	x	0	1	1
0	0	0	1	x	x	x	x	1	0	0
0	0	1	x	x	x	x	x	1	0	1
0	1	x	x	x	x	x	x	1	1	0
1	x	x	x	x	x	x	x	1	1	1

$$Q_2 = \bar{D}_7 \bar{D}_6 \bar{D}_5 D_4 + \bar{D}_7 \bar{D}_6 D_5 + \bar{D}_7 D_6 + D_7$$

$$Q_2 = \bar{D}_7 \bar{D}_6 \bar{D}_5 D_4 + \bar{D}_7 \bar{D}_6 D_5 + D_6 + D_7$$

$$Q_2 = \bar{D}_7 \bar{D}_6 \bar{D}_5 D_4 + D_5 \overline{(D_6 + D_7)} + (D_6 + D_7)$$

$$Q_2 = \bar{D}_7 \bar{D}_6 \bar{D}_5 D_4 + D_5 + D_6 + D_7$$

$$Q_2 = D_4 + D_5 + D_6 + D_7.$$

$$Q_2 = D_4 + D_5 + D_6 + D_7.$$

$Q_2 = D_4 +$

$Q_1 = \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,\overline{D_3}\,D_2 + \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,D_3 + \overline{D_7}\,D_6 + D_7.$

$Q_1 = \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,\overline{D_3}\,D_2 + \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,D_3 + D_6 + D_7$

$Q_1 = \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,\overline{D_3}\,D_2 + \overline{D_5}\,\overline{D_4}\,D_3 + D_6 + D_7$

$Q_1 = \overline{D_5}\,\overline{D_4}\,\overline{D_3}\,D_2 + \overline{D_5}\,\overline{D_4}\,D_3 + D_6 + D_7.$

$Q_1 = \overline{D_5}\,\overline{D_4}\,(D_2 + D_3) + D_6 + D_7.$

$Q_1 = \overline{D_5}\,\overline{D_4}\,D_3 + \overline{D_5}\,\overline{D_4}\,D_2 + D_6 + D_7.$

$Q_0 = \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,\overline{D_3}\,\overline{D_2}\,D_1 + \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,D_3 + \overline{D_7}\,\overline{D_6}\,D_5 + D_7.$

$Q_0 = \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,\overline{D_3}\,\overline{D_2}\,D_1 + \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,D_3 + \overline{D_6}\,D_5 + D_7$

$Q_0 = \overline{D_7}\,\overline{D_6}\,\overline{D_5}\,\overline{D_4}\,\overline{D_3}\,\overline{D_2}\,D_1 + \overline{D_6}\,\overline{D_5}\,\overline{D_4}\,D_3 + \overline{D_6}\,D_5 + D_7.$

$Q_0 = \overline{D_6}\left(\overline{D_5}\,\overline{D_4}\,\overline{D_3}\,\overline{D_2}\,D_1 + \overline{D_5}\,\overline{D_4}\,D_3 + D_5\right) + D_7.$

$Q_0 = \overline{D_6}\left(\overline{D_4}\,\overline{D_3}\,\overline{D_2}\,D_1 + \overline{D_4}\,D_3 + D_5\right) + D_7$

$Q_0 = \overline{D_6}\left(\overline{D_4}\left(\overline{D_2}\,D_1 + D_3\right) + D_5\right) + D_7.$

$Q_0 = \overline{D_6}\,\overline{D_4}\,\overline{D_2}\,D_1 + \overline{D_6}\,\overline{D_4}\,D_3 + \overline{D_6}\,D_5 + D_7.$